

# DW\_sincos

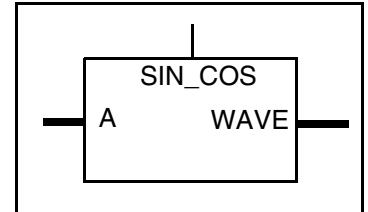
## Sine and Cosine

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### Features and Benefits

- Parameterized word length
- sine or cosine output by controlling SIN\_COS port.
- DesignWare Datapath generator is employed for better timing and area.

### Revision History



### Description

DW\_sincos is a fixed point sine and cosine unit that calculates  $WAVE = \sin(\pi A)$  or  $\cos(\pi A)$  where  $A$  and  $WAVE$  are fixed point value by controlling the  $SIN\_COS$  pin. DW\_sincos has two additional parameters: *arch* is for the implementation between the area-optimized and the speed-optimized, and *err\_range* allows users to choose the error range between 1 ulp error and 2 ulp error.

**Table 1-1 Pin Description**

| Pin Name | Width              | Direction | Function                                                                                                            |
|----------|--------------------|-----------|---------------------------------------------------------------------------------------------------------------------|
| A        | $A\_width$ bits    | Input     | Input data (radian)                                                                                                 |
| SIN_COS  | 1 bit              | Input     | Function select: <ul style="list-style-type: none"> <li>■ 0: Sine function</li> <li>■ 1: Cosine function</li> </ul> |
| WAVE     | $WAVE\_width$ bits | Output    | Sine or cosine value                                                                                                |

**Table 1-2 Parameter Description**

| Parameter  | Values               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A_width    | 3 to 34 bits         | Word length of $A$                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| WAVE_width | 2 to 35 bits         | Word length of $WAVE$                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| arch       | 0 or 3<br>Default: 0 | Select the implementation <ul style="list-style-type: none"> <li>■ 0: Area-optimized implementation (for behavior that is backward-compatible with DC releases up to R-2020.09)</li> <li>■ 1: Speed-optimized implementation (for behavior that is backward-compatible with DC releases up to R-2020.09)</li> <li>■ 2: Area-optimized implementation with <i>err_range</i> control</li> <li>■ 3: Speed-optimized implementation with <i>err_range</i> control</li> </ul> |

**Table 1-2 Parameter Description (Continued)**

| Parameter | Values               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| err_range | 1 or 2<br>Default: 1 | Select the error range<br>For <i>arch</i> = 0 or 1: <ul style="list-style-type: none"><li>The error bound is always 2 ulp, regardless of the <i>err_range</i> value (this behavior is preserved to ensure backward-compatibility with DC releases up to R-2020.09)</li></ul> For <i>arch</i> = 2 or 3, the following values select the error range: <ul style="list-style-type: none"><li>1: <math> \text{true value} - \text{calculated}  &lt; 1 \text{ ulp}^{\text{ab}}</math></li><li>2: <math> \text{true value} - \text{calculated}  &lt; 2 \text{ ulp}^{\text{ab}}</math></li></ul> |

a. 1 ulp corresponds to the weight of the LSB of the output value, which is equivalent to  $2^{-(\text{WAVE\_width}+2)}$ .

b. When *arch* = 2 or 3 and the precision is 25 bits or more, the error can reach 3 ulp.

**Table 1-3 Synthesis Implementations**

| Implementation Name | Function        | License Feature Required |
|---------------------|-----------------|--------------------------|
| rtl                 | Synthesis model | DesignWare               |

**Table 1-4 Simulation Models**

| Model                         | Function                             |
|-------------------------------|--------------------------------------|
| DW02.DW_SINCOS_CFG_SIM        | Design unit name for VHDL simulation |
| dw/dw02/src/DW_sincos_sim.vhd | VHDL simulation model source code    |
| dw/sim_ver/DW_sincos.v        | Verilog simulation model source code |

## Error Behavior

The parameter *err\_range* controls the error level at the *WAVE* output. When *err\_range* = 1, the *WAVE* value has an error of at most  $2^{-(\text{WAVE\_width}+2)}$ , which is the weight of the *WAVE* LSB representation. For example, with *WAVE\_width* = 6 and the long-precision calculation of the function is 01.100011, the output can be 01.1000 or 01.1001. The error is  $2^{-(\text{WAVE\_width}+2)} = 2^{-4}$ .

When *err\_range* = 2, the value of *WAVE* may have a larger error up to  $2^{-(\text{WAVE\_width}+3)}$ . By increasing the error bound, you can reduce the hardware complexity, save area, and reduce the critical path.

**Note**

As explained in [Table 1-2](#), the *err\_range* parameter has no effect on the hardware complexity or numerical behavior when *arch* = 0 or *arch* = 1. In these configurations, the error range varies depending on the configuration.

## Functional Description

The input angle  $A$  is treated as a binary fixed point number which is converted to radians when multiplied by  $\pi$ . When  $A$  is interpreted as unsigned, the input angle  $A$  is a binary subdivision of the range  $0 \leq A < 2$ . When  $A$  is interpreted as signed (two's complement), the range is  $-1 \leq A < 1$ .

**Table 1-5 Operations of DW\_sincos**

| SIN_COS input | A input                            | WAVE output   |
|---------------|------------------------------------|---------------|
| 0             | $A$ (unsigned or two's complement) | $\sin(\pi a)$ |
| 1             | $A$ (unsigned or two's complement) | $\cos(\pi a)$ |

## Input Data Format

The input value  $a$  is interpreted with the MSB as the only bit to the left of the decimal point. For example, if  $A\_width = 6$ , the input value 111000 is interpreted as 1.11000.

The sine (or cosine) value calculated by DW\_sincos is the same whether  $A$  is interpreted as an unsigned number or as a signed (two's complement) number. For example, consider the case where  $SIN\_COS = 0$  and the input value  $A$  is 1.10000. Interpreted as an unsigned number, the decimal equivalent of  $A$  is  $3/2$ , the angle is  $3\pi/2$ , and the sine value is -1 (decimal).

When 1.10000 is interpreted as a signed number, the MSB becomes the sign bit (1 in this example, indicating a negative number) and .10000 is the two's complement of .10000. The decimal equivalent is  $(-1/2)$ . DW\_sincos calculates the sine of  $(-\pi/2)$ , which is also -1 (decimal).

Similarly, when  $SIN\_COS = 1$ , DW\_sincos calculates the cosine. If the input value ( $A = 1.10000$ ) is interpreted as an unsigned number, DW\_sincos calculates the cosine of  $3\pi/2$ , which equals 0. If the input value ( $A = 1.10000$ ) is interpreted as a two's complement number, DW\_sincos calculates the cosine of  $(-\pi/2)$ , which also equals 0.

## Output Data Format

The output value  $WAVE$  is of the form  $SX.XXXX$ , where  $S$  is the sign bit and  $X.XXXX$  is the two's complement of the binary value of the sine or cosine. For example, the sine value 1 (decimal) is represented as 01.0000 (when  $WAVE\_width = 6$ ), and the sine value -1 is represented as 11.0000.

Table 1-6 shows the bit assignments for the input and output pins. Table 7 shows input and output values for an example implementation where  $A\_width = 6$ ,  $WAVE\_width = 6$ .

**Table 1-6 Input and Output Data Format**

| Pin                     | Range (binary)                                                                                                                                      | Range (decimal)       |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| A (unsigned)            | 1.111...111 ( $< 2$ )<br>1.111...110<br>...<br>1.000...000 ( $= 1$ )<br>...<br>0.000...001<br>0.000...000 ( $= 0$ )                                 | $0 \leq A < 2$        |
| A (two's complement)    | 0.111...111 ( $< 1$ )<br>0.111...110<br>...<br>0.000...000 ( $= 0$ )<br>1.111...111 ( $< 0$ )<br>...<br>1.000...001<br>1.000...000 ( $= -1$ )       | $-1 \leq A < 1$       |
| WAVE (two's complement) | 01.000...000 ( $= 1$ )<br>00.111...111<br>...<br>00.000...000 ( $= 0$ )<br>11.111...111 ( $< 0$ )<br>...<br>11.000...001<br>11.000...000 ( $= -1$ ) | $-1 \leq WAVE \leq 1$ |

**Table 1-7 Sample Data Value for  $A\_width = 6$  and  $WAVE\_width = 6$**

| Value of A                 | SIN_COS | Function       | Value of WAVE |
|----------------------------|---------|----------------|---------------|
| 1.00000 (unsigned)         | 0       | $\sin(\pi)$    | 00.0000       |
| 1.10000 (unsigned)         | 0       | $\sin(3\pi/2)$ | 11.0000       |
| 1.00000 (two's complement) | 1       | $\cos(-\pi)$   | 11.0000       |
| 0.10000 (two's complement) | 0       | $\sin(\pi/2)$  | 01.0000       |

## Alternative Implementation of Sine and Cosine with DW\_lp\_multifunc

The sine and cosine operation can also be implemented by DW\_lp\_multifunc component, which evaluates the value of sine and cosine with an error bounded by the weight of the LSB of the output WAVE. The value from DW\_lp\_multifunc and the value from DW\_sincos may differ by at most 1 LSB. Performance and area of the synthesis results are different between the DW\_sincos and sine and cosine implementation of the DW\_lp\_multifunc, depending on synthesis constraints, library cells and synthesis environments. By comparing performance and area between the sine and cosine implementation of DW\_lp\_multifunc and DW\_sincos component, the DW\_lp\_multifunc provides more choices for the best synthesis results. Below is an example of the Verilog description for the sine and cosine of the DW\_lp\_multifunc. For more detailed information, see the [DW\\_lp\\_multifunc](#) datasheet.

```
DW_lp_multifunc #(op_width, 24) U1 (  
    .a(a),  
    .func(func),  
    .z(z),  
    .status(status)  
);
```

## Related Topics

- [Datapath – Floating Point Overview](#)
- [DesignWare Building Block IP User Guide](#)

## HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_Foundation_comp_arith.all;

entity DW_sincos_inst is
    generic (
        inst_A_width : INTEGER := 24;
        inst_WAVE_width : INTEGER := 25;
        inst_arch : INTEGER := 0;
        inst_err_range : INTEGER := 1
    );
    port (
        inst_A : in std_logic_vector(inst_A_width-1 downto 0);
        inst_SIN_COS : in std_logic;
        WAVE_inst : out std_logic_vector(inst_WAVE_width-1 downto 0)
    );
end DW_sincos_inst;

architecture inst of DW_sincos_inst is

begin

    -- Instance of DW_sincos
    U1 : DW_sincos
    generic map (
        A_width => inst_A_width,
        WAVE_width => inst_WAVE_width,
        arch => inst_arch,
        err_range => inst_err_range
    )
    port map (
        A => inst_A,
        SIN_COS => inst_SIN_COS,
        WAVE => WAVE_inst
    );

end inst;
```

## HDL Usage Through Component Instantiation - Verilog

```
module DW_sincos_inst( inst_A, inst_SIN_COS, WAVE_inst );

parameter inst_A_width = 24;
parameter inst_WAVE_width = 25;
parameter inst_arch = 0;
parameter inst_err_range = 1;

input [inst_A_width-1 : 0] inst_A;
input inst_SIN_COS;
output [inst_WAVE_width-1 : 0] WAVE_inst;

// Instance of DW_sincos
DW_sincos #(inst_A_width, inst_WAVE_width, inst_arch, inst_err_range) U1 (
    .A(inst_A),
    .SIN_COS(inst_SIN_COS),
    .WAVE(WAVE_inst) );

endmodule
```

## Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

| Date           | Release       | Updates                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| September 2024 | DWBB_202409.0 | <ul style="list-style-type: none"> <li>Updated date and version; no technical changes</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| December 2023  | DWBB_202312.0 | <ul style="list-style-type: none"> <li>Updated date and version; no technical changes</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| October 2020   | DWBB_202009.1 | <ul style="list-style-type: none"> <li>For STAR 9001562033:               <ul style="list-style-type: none"> <li>In <a href="#">Table 1-2</a> on page 1:                   <ul style="list-style-type: none"> <li>Updated the lower range of A_width</li> <li>Added new architecture values for <i>arch</i></li> <li>Updated the description of <i>err_range</i></li> </ul> </li> <li>Added <a href="#">“Error Behavior”</a> on page 2</li> <li>Clarified details in <a href="#">“Alternative Implementation of Sine and Cosine with DW_lp_multifunc”</a> on page 5</li> </ul> </li> </ul> |
| January 2020   | DWBB_201912.1 | <ul style="list-style-type: none"> <li>Corrected port names for DW_lp_multifunc in <a href="#">“Alternative Implementation of Sine and Cosine with DW_lp_multifunc”</a> on page 5</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                               |
| July 2019      | DWBB_201903.3 | <ul style="list-style-type: none"> <li>Removed reference to minPower library in <a href="#">“Alternative Implementation of Sine and Cosine with DW_lp_multifunc”</a> on page 5</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                  |
| March 2019     | DWBB_201903.0 | <ul style="list-style-type: none"> <li>Removed minPower designation from this datasheet</li> <li>Added this Revision History table and the document links on this page</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                          |

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